



Name:

Mathematics Class:

Year 12

Mathematics Extension 1

HSC Course

Assessment 4 - Trial

August 2019

Time allowed: 120 minutes + 5 minutes reading time

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A reference sheet is provided

Section 1 Multiple Choice

Questions 1-10

10 Marks

Section II Questions 11-14

60 Marks

Section 1

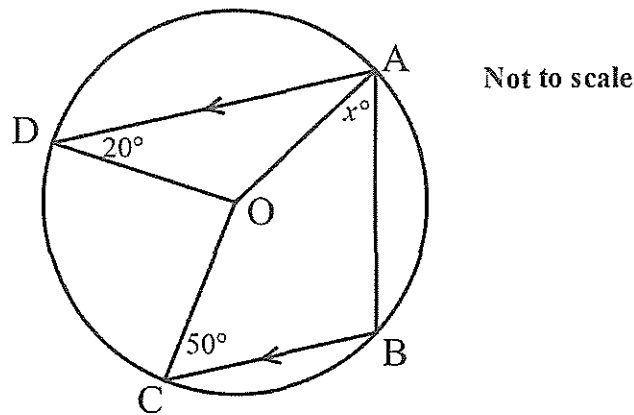
Multiple Choice (10 marks)

Attempt Questions 1 - 10

Use the multiple-choice answer sheet for Questions 1-10

1. What is the value of $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{1}{2}x\right)}{2x}$?
- A. 0
- B. $\frac{1}{4}$
- C. 1
- D. 4
2. What are the coordinates of the point P that divides the interval joining the points A(1,2) and B(7,5) internally in the ratio 2:1?
- A. (3,3)
- B. (3,4)
- C. (5,4)
- D. (5,3)
3. When the polynomial $P(x) = x^3 - 5x^2 + kx + 2$ is divided by $(x + 1)$ the remainder is 3. What is the value of k ?
- A. -7
- B. -5
- C. 5
- D. 7

4. A, B, C and D are points on a circle with centre O . BC is parallel to AD .
 $\angle ADO = 20^\circ$ and $\angle BCO = 50^\circ$. Let $\angle BAO = x^\circ$.

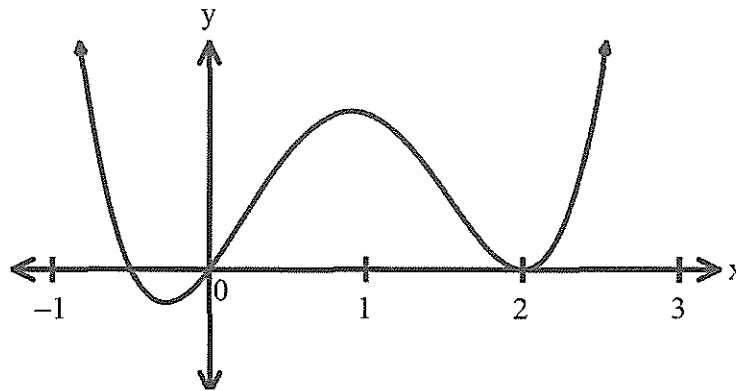


What is the value of x ?

- A. 15
 B. 35
 C. 40
 D. 55
5. Which of the following is a simplification of $4 \log_e(\sqrt{e^x})$?
- A. $4\sqrt{x}$
 B. $\frac{1}{2}x$
 C. $2x$
 D. x^2
6. The acute angle between the lines $2x - y = 0$ and $kx - y = 0$ is equal to $\frac{\pi}{4}$.
 —What is the value of k ?

- A. $k = -3$ or $k = -\frac{1}{3}$
 B. $k = -3$ or $k = \frac{1}{3}$
 C. $k = 3$ or $k = -\frac{1}{3}$
 D. $k = 3$ or $k = \frac{1}{3}$

7. The graph of a polynomial function is shown below.



What could be the equation of the polynomial represented by the graph?

- A. $y = x(2x - 1)(x - 1)^2$
- B. $y = x(2x - 1)(x - 2)^2$
- C. $y = x(2x + 1)(x - 2)^2$
- D. $y = x(2x + 1)(x + 1)^2$
8. Which one of following is an expression for $\int \frac{3}{\sqrt{1 - 16x^2}} dx$?

- A. $3\sin^{-1}(4x) + C$
- B. $3\cos^{-1}(4x) + C$
- C. $\frac{3}{4}\sin^{-1}(4x) + C$
- D. $\frac{3}{4}\cos^{-1}(4x) + C$

9. Which is an expression for $\frac{d}{dx}(\tan^{-1}(2x + 1))$?

A. $\frac{1}{4x^2 + 4x + 2}$

B. $\frac{1}{2x^2 + 2x + 1}$

C. $\frac{1}{4x^2 + 2}$

D. $\frac{1}{2x^2 + 1}$

10. What is the general solution to the equation $2\sin^2\theta + 5\cos\theta + 1 = 0$

A. $2n\pi \pm \frac{2\pi}{3}$ where n is an integer.

B. $2n\pi \pm \frac{5\pi}{6}$ where n is an integer.

C. $2n\pi \pm \frac{\pi}{6}$ where n is an integer.

D. $2n\pi \pm \frac{\pi}{3}$ where n is an integer.

Section II

Total Marks (60)

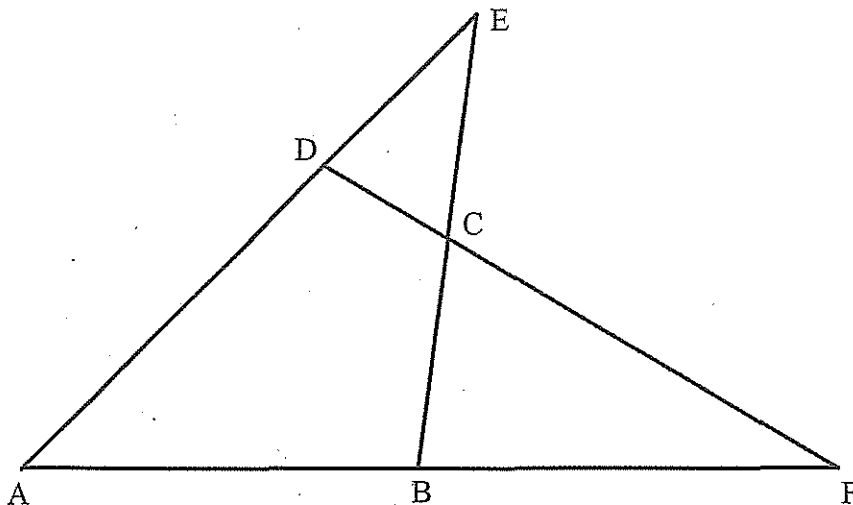
Attempt Questions 11 – 14.

Answer each question in your writing booklet.

In Questions 11 - 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 mark)

- (a) Solve $\frac{3}{4x-1} \geq 2$ 3
- (b) Solve the equation $\sin 2x + \cos x = 0$ for $0 \leq x \leq 2\pi$ 3
- (c) Find the exact value of $\sin \left[\cos^{-1} \left(\frac{2}{3} \right) + \tan^{-1} \left(-\frac{3}{4} \right) \right]$. 3
- (d) Differentiate $x^2 \cos^{-1}(2x)$. 2
- (e) In the diagram $ABCD$ is a cyclic quadrilateral. AD produced and BC produced meet at E . AB produced and DC produced meet at F . $\angle DEC = \angle BFC$.



- (i) Copy the diagram into your answer booklet.
- (ii) Show that $\angle ADC = \angle ABC$. 2
- (ii) Show that AC is a diameter of the circle through A, B, C and D 2

End of Question 11

Question 12 (15 mark) (start a new page)

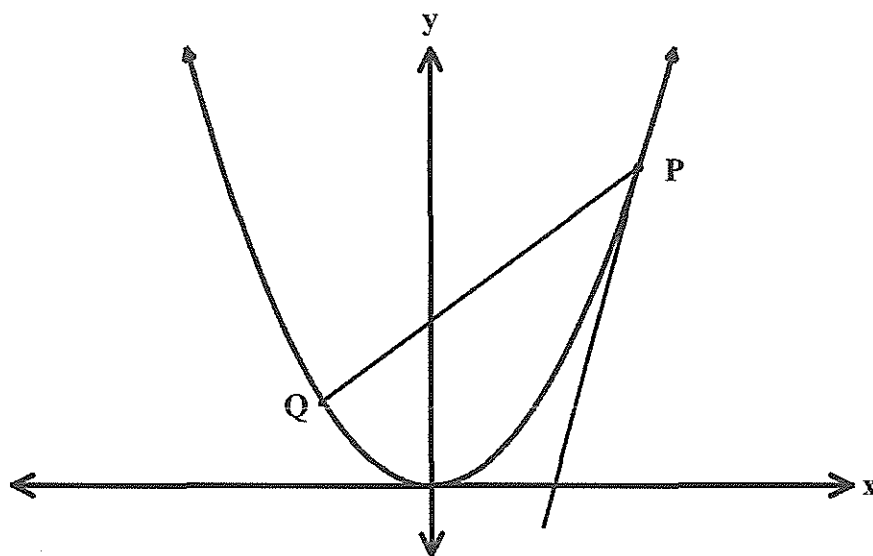
- (a) Use Mathematical Induction to show that for all positive integers $n \geq 1$ 3

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n.$$

- (b) An oil slick in the shape of a circle is spreading across a lake, such that its radius is increasing at a rate of 0.1 m/s.

Find the radius of the oil slick when its area is increasing at a rate of $2\pi m^2/s$.

2



(c)

The diagram above shows the variable points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ on the parabola $x^2 = 4ay$.

M is the midpoint of PQ .

P and Q move such that the gradient of the tangent at P , is four times the gradient of the chord PQ .

- (i) Show that $q = -\frac{1}{2}p$ 2

- (ii) Show that as p varies, M moves on a parabola. 2

(d) Use the substitution $u = x + 1$ to evaluate in simplest exact form $\int_2^5 \frac{x+2}{(x+1)^2} dx$. 3

(e) The region bounded by the curve $y = \cos^{-1}x$ and the y axis between $y = \frac{\pi}{12}$ and $y = \frac{\pi}{4}$ is rotated through one complete revolution about the y-axis. Find the exact volume of the solid formed. 3

End of Question 12

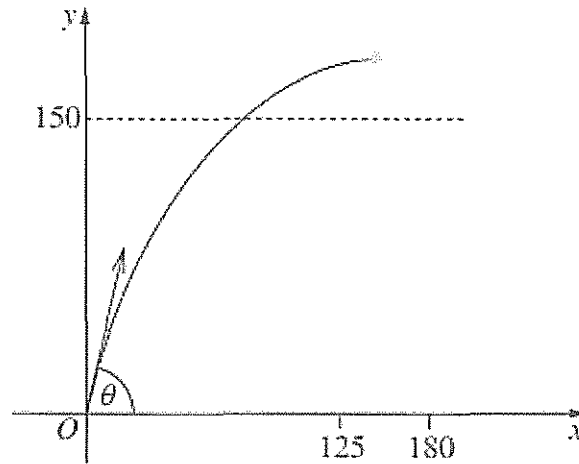
Question 13 (15 mark) (start a new page)

- (a) At time t years the number N of individuals in a population is such that $N = P + 100e^{kt}$ for some constants P and k .
- (i) Show that $\frac{dN}{dt} = k(N - P)$. 1
- (ii) If initially the population size is 600 and is increasing at a rate of 12 individuals per year, find the values of P and k . 2
- (b) On a certain day, low tide for a harbour occurs at 4:00 am and high tide occurs at 10:20 am. The corresponding depths are 3 m and 5 m respectively. The tidal motion is assumed to be simple harmonic.
- (i) Show that the water depth, y metres, is given by $y = 4 - \cos \frac{3\pi t}{19}$ where t is the number of hours after low tide. 2
- (ii) A boat requires a depth of at least 4.5m. What is the earliest and the latest time that the boat can be in the harbour before 5:00pm on that day? 2
- (c) A particle is moving in a straight line. At time t seconds it has displacement x metres to the right of a fixed point O on the line and velocity $v \text{ ms}^{-1}$ given by $v = \frac{16 - x^2}{x}$. Initially the particle is 1 metre to the right of O .
- (i) Show that $x = \sqrt{16 - 15e^{-2t}}$. 3
- (ii) Find the limiting position of the particle. 1
- (d) Consider the function $f(x) = \sin^{-1}(1 - x) + \frac{\pi}{2}$.
- (i) Find the domain and range of the function. 2
- (ii) Sketch the graph of the function, clearly showing the shape of the curve and the co-ordinates of the endpoints. 2

End of Question 13

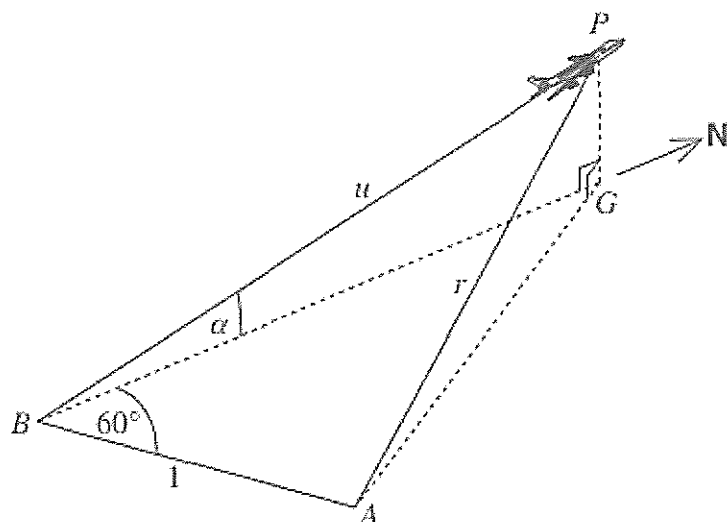
Question 14 (15 mark) (start a new page)

- (a) A firework is fired from O, on level ground, with velocity 70 metres per second at an angle of inclination θ . The equations of motion of the firework are $x = 70t \cos\theta$ and $y = 70t \sin\theta - 4.9t^2$. (Do NOT prove this.)
The firework explodes when it reaches its maximum height.



- (i) Show that the firework explodes at a height of $250 \sin^2\theta$ metres 2
- (ii) Show that the firework explodes at a horizontal distance of $250 \sin 2\theta$ metres from O. 1
- (iii) For best viewing, the firework must explode at a horizontal distance between 125 m and 180 m from O, and at least 150 m above the ground. For what values of θ will this occur? 3
- (b) (i) Show that $1 + x + x^2 + x^3 + \dots + x^n = \frac{x^{n+1} - 1}{x - 1}$ 1
- (ii) Use the result from b(i) to show that
- $$1 \times 1 + 2 \times 2 + 3 \times 4 + 4 \times 8 + \dots + n \times 2^{n-1} = 2^n(n-1) + 1$$
- 3

- (c) A plane P takes off from a point B . It flies due north at a constant angle α to the horizontal. An observer is located at A , 1 km from B , at a bearing 060° from B . Let u km be the distance from B to the plane and let r km be the distance from the observer to the plane. The point G is on the ground directly below the plane.



- (i) Show that $r = \sqrt{1 + u^2 - u \cos \alpha}$. 3
- (ii) The plane is travelling at a constant speed of 360 km/h. At what rate, in terms of α , is the distance of the plane from the observer changing 5 minutes after take-off? 2

End of Examination



Question 14:

a) Please tell the examiner WHY you are setting $\dot{y} = 0$

a) (iii) Most students used their information found in parts (i) and (ii) to do this part.

If so, then one set of solutions gives that $\theta \geq 51^\circ$. The other set of solutions sees the solution to

$\frac{1}{2} \leq \sin 2\theta \leq \frac{18}{25}$. There are 2 sets of answers here: the first is in the first quadrant, and does not give angles greater than 51° so offers no solution. Most students somehow got to this point. BUT there are a second set of solutions in quadrant 2, leading to $67^\circ \leq \theta \leq 75^\circ$. Not many were perceptive enough to get these.

The other "method" was to find the equation of the trajectory, then substitute in $y = 150$ and $x = 125$ then $x = 180$, and solving for θ . This was very involved and led to many errors. Probably two students who did this 2 pages of work involved got anywhere.

b) (i) The formula is given. You cannot just "requote it"! The important part is that there are $n+1$ terms.

There were 3 methods:

1. The one in your solutions.
2. Multiply $(1 + x + x^2 + x^3 + \dots)$ by $x - 1$
3. Let $S_n = 1 + x + x^2 + x^3 + \dots + x^n$

$$\text{then } x.S_n = x + x^2 + x^3 + \dots + x^n + x^{n+1}$$

$$\text{So } S_n(x - 1) = x^{n+1} - 1$$

$$\rightarrow S_n = 1 + x + x^2 + x^3 + \dots + x^n = \frac{x^{n+1} - 1}{x - 1}$$

For 1 mark, don't start using Mathematical Induction!

(b) (ii) Using Mathematical Induction was NOT following instructions to use part (b) (i) above., but would get you 2 marks out of 3 if it was done well (not often!)

(c) Last question, so meant to be difficult.

Year 12 Extension 1 Trial - 2019

Section 1

1 B

2 C

3 A

4 D

5 C

6. B

7 C

8 C

9. B

10. A

1. $\lim_{x \rightarrow 0} \frac{\sin(\frac{1}{2}x)}{2x}$

$$= \frac{1}{4} \lim_{x \rightarrow 0} \frac{\sin(\frac{1}{2}x)}{\frac{1}{2}x}$$

$$= \frac{1}{4}$$

B

2. (1, 2) (7, 5) 2:1

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$x = \frac{2 \times 7 + 1 \times 1}{3}$$

$$y = \frac{2 \times 5 + 1 \times 2}{3}$$

$$x = 5$$

$$y = 4$$

C

3. $P(-1) = 3$

$$x^3 - 5x^2 + kx + 2 = (-1)^3 - 5(-1) + k(-1) + 2$$

$$3 = 1 - 5 - k + 2$$

$$3 = -4 - k$$

$$k = -7$$

A

4. Construct $DF \parallel AD$

Join O to B

$$\angle DOC = 20^\circ + 50^\circ = 70^\circ$$

$$\angle DOA = 140^\circ$$

$$\angle COB = 80^\circ$$

$$\angle BOA = 70^\circ$$

$$2x + 70^\circ = 180^\circ$$

$$x = 55^\circ$$

D

5.

$$\begin{aligned} & 4 \log_e \sqrt{e^x} \\ &= 4 \log_e e^{\frac{x}{2}} \\ &= 4 \times \frac{1}{2} \log_e e^x \\ &= 4 \times \frac{1}{2} x \\ &= 2x \end{aligned}$$

C

6. $\tan \frac{\pi}{4} = 1$

$$1 = \left| \frac{k-2}{1+2k} \right|$$

$$1+2k = k-2$$

or

$$1+2k = -(k-2)$$

$$k = -3$$

$$3k = 1$$

$$k = \frac{1}{3}$$

B

7. $(x-2)^2$ double root

x single root

root in between -1 and 0.

C

$$\begin{aligned}
 8 \quad \int \frac{3}{\sqrt{1-16x^2}} dx &= 3 \int \frac{1}{\sqrt{1-16x^2}} dx \\
 &= 3 \int \frac{1}{\sqrt{16(\frac{1}{16}-x^2)}} dx \\
 &= \frac{3}{4} \sin^{-1}(4x) + C
 \end{aligned}$$

$$\begin{aligned}
 9. \quad \frac{d}{dx} \tan^{-1}(2x+i) &= \frac{2}{1+(2x+i)^2} \\
 &= \frac{2}{4x^2+4x+2} \\
 &= \frac{1}{2x^2+2x+1}
 \end{aligned}$$

$$\begin{aligned}
 10 \quad 2\sin^2\theta + 5\cos\theta + 1 &= 0 \\
 2(1-\cos^2\theta) + 5\cos\theta + 1 &= 0 \\
 2 - 2\cos^2\theta + 5\cos\theta + 1 &= 0 \\
 2\cos^2\theta - 5\cos\theta - 3 &= 0 \\
 (2\cos\theta + 1)(\cos\theta - 3) &= 0 \\
 \cos\theta = -\frac{1}{2} &\quad \uparrow \text{reject} \\
 \theta &= 2n\pi \pm \cos^{-1}\left(-\frac{1}{2}\right) \\
 &= 2n\pi \pm \frac{2\pi}{3}
 \end{aligned}$$

Question 11

$$a) \frac{3}{4x-1} \geq 2$$

$$3(4x-1) \geq 2(4x-1)^2$$

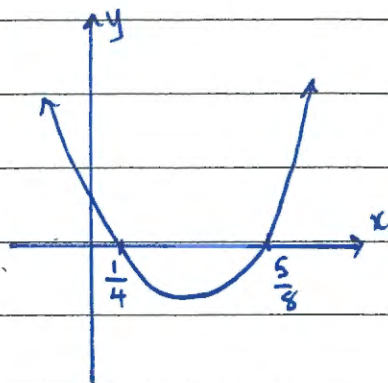
$$0 \geq 2(4x-1)^2 - 3(4x-1)$$

$$0 \geq (4x-1)[2(4x-1)-3]$$

$$0 \geq (4x-1)(8x-2-3)$$

$$0 \geq (4x-1)(8x-5)$$

$$\frac{1}{4} < x \leq \frac{5}{8}$$



$$b) \sin 2x + \cos x = 0$$

$$2 \sin x \cos x + \cos x = 0$$

$$\cos x (2 \sin x + 1) = 0$$

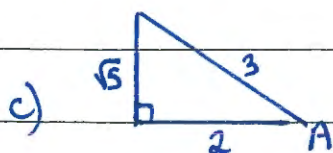
$$\cos x = 0$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$2 \sin x + 1 = 0$$

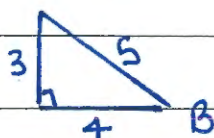
$$\sin x = -\frac{1}{2}$$

$$x = \frac{7\pi}{6}, \frac{11\pi}{6}$$



$$\cos A = \frac{2}{3}$$

$$\sin A = \frac{\sqrt{5}}{3}$$



$$B = \tan^{-1} \frac{3}{4}$$

$$\tan B = \frac{3}{4}$$

$$\sin B = \frac{3}{5}$$

$$\cos B = \frac{4}{5}$$

$$= \sin \left[\cos^{-1} \left(\frac{2}{3} \right) + \tan^{-1} \left(-\frac{3}{4} \right) \right]$$

$$= \sin \left[\cos^{-1} \frac{2}{3} - \tan^{-1} \left(\frac{3}{4} \right) \right]$$

$$\sin(A-B)$$

$$= \sin A \cos B - \cos A \sin B$$

$$= \frac{\sqrt{5}}{3} \times \frac{4}{5} - \frac{2}{3} \times \frac{3}{5}$$

$$= \frac{4\sqrt{5} - 6}{15}$$

$$d) \quad u = x^2 \quad v = \cos^{-1}(2x)$$

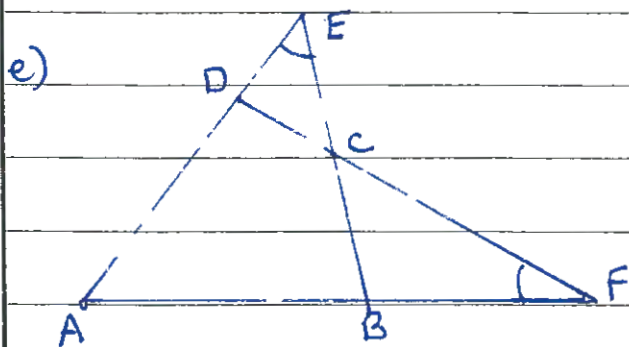
$$du = 2x \, dx$$

$$dv = \frac{-1}{\sqrt{1-(2x)^2}} \times 2$$

$$\frac{d}{dx} (x^2 \cos^{-1}(2x))$$

$$= \cos^{-1}(2x) \times 2x + x^2 \times \frac{-1}{\sqrt{1-(2x)^2}} \times 2$$

$$= 2x \cos^{-1}(2x) - \frac{2x^2}{\sqrt{1-4x^2}}$$



In $\triangle DEC$

$$\angle ADC = \angle DEC + \angle ECD$$

exterior angle is equal to the sum of the ^{two} opposite interior angles.

In $\triangle BFC$

$\angle ABC = \angle BFC + \angle FCB$ (exterior angle is equal to the sum of the two opposite interior angle)

$\angle DCE = \angle BCF$ (vertically opposite angles equal)

$$\therefore \angle ADC = \angle ABC$$

e ii) $\angle ADC + \angle ABC = 180^\circ$

(opposite angles of a cyclic quadrilateral are supplementary)

$$\therefore \angle ABC = 90^\circ$$

$\therefore AC$ is diameter of circle $ABCD$

Question 12

a) $S_n = 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n$

Show true for $n=1$

$$L.H.S = 1 \times 2^{1-1}$$

$$= 1$$

$$R.H.S = 1 + (1-1)2^1$$

$$= 1$$

$\therefore$ true for $S(1)$

Assume true for $n=k$

ie $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1)2^k$

Prove true for $n=k+1$

ie prove $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} + (k+1)2^k = 1 + (k+1)2^{k+1}$

$$L.H.S = 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} + (k+1) \times 2^k$$

$$= 1 + (k-1)2^k + (k+1)2^k$$

$$= 1 + (k-1+k+1)2^k$$

$$= 1 + 2k \cdot 2^k$$

$$= 1 + [(k+1)-1]2^{k+1}$$

$$= R.H.S$$

If $S(k)$ is true then $S(k+1)$ is true, $S(1)$ is true, $S(2)$ is true and then $S(3)$ is true and so on.

$\therefore S(n)$ is true for all positive integers $n \geq 1$

$$b) \quad A = \pi r^2$$

$$\frac{dr}{dt} = 0.1$$

$$\frac{dA}{dt} = 2\pi$$

$$r = ?$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$2\pi = 2\pi r \times 0.1$$

$$0.1r = 1$$

$$r = 10$$

$\therefore$ Radius is 10m when the area is increasing at $2\pi \text{ m}^2/\text{s}$

$$ci) \quad x^2 = 4ay$$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

$$\text{At } P(2ap, ap^2)$$

$$M_{\text{tangent}} = \frac{2ap}{2a} = p$$

$$M_{\text{chord}} = \frac{ap^2 - aq^2}{2ap - 2aq}$$

$$= \frac{a(p-q)(p+q)}{2a(p-q)}$$

$$= \frac{p+q}{2}$$

$$M_{\text{tangent}} = 4 \times M_{\text{chord}}$$

$$p = 4 \times \frac{p+q}{2}$$

$$p = 2p + 2q$$

$$-p = 2q$$

$$\therefore q = -\frac{1}{2}p \text{ as required}$$

$$\text{c ii)} \quad M = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right) \quad - (1)$$

$$q = -\frac{1}{2} p \quad - (2)$$

$$(2) \rightarrow (1)$$

$$M = \left(\frac{2ap + 2a\left(-\frac{p}{2}\right)}{2}, \frac{ap^2 + a\left(-\frac{p}{2}\right)^2}{2} \right)$$

$$= \left(\frac{2ap - ap}{2}, \frac{ap^2 + \frac{ap^2}{4}}{2} \right)$$

$$= \left(\frac{ap}{2}, \frac{5ap^2}{8} \right)$$

$$\text{Locus of } M = \quad x = \frac{ap}{2} \quad p = \frac{2x}{a}$$

$$y = \frac{5ap^2}{8}$$

$$y = \frac{5a}{8} \times p^2$$

$$y = \frac{5a}{8} \times \left(\frac{2x}{a} \right)^2$$

$$y = \frac{5a}{8} \times \frac{4x^2}{a^2}$$

$$y = \frac{5x^2}{2a}$$

$2ay = 5x^2$ which is the form of a parabola

$\therefore$ M moves on a parabola.

$$d) \quad u = x + 1 \quad x = 2 \Rightarrow u = 3$$

$$du = 1 \quad x = 5 \Rightarrow u = 6$$

$$\int_2^5 \frac{x+2}{(x+1)^2} dx = \int_3^6 \frac{u+1}{u^2} du$$

$$= \int_3^6 \frac{1}{u} + \frac{1}{u^2} du$$

$$= \left[\ln u - \frac{1}{u} \right]_3^6$$

$$= (\ln 6 - \frac{1}{6}) - (\ln 3 - \frac{1}{3})$$

$$= (\ln 6 - \ln 3) - (\frac{1}{6} - \frac{1}{3})$$

$$= \ln 2 + \frac{1}{6}$$

$$e) \quad y = \cos^{-1} x$$

$$x = \cos y$$

$$V = \pi \int_{\pi/12}^{\pi/4} \cos^2 y \, dy$$

$$= \frac{\pi}{2} \int_{\pi/12}^{\pi/4} (1 + \cos 2y) \, dy$$

$$= \frac{\pi}{2} \left[y + \frac{1}{2} \sin 2y \right]_{\pi/12}^{\pi/4}$$

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{4} - \frac{\pi}{12} \right) - \frac{1}{2} \left(\sin \frac{\pi}{2} - \sin \frac{\pi}{6} \right) \right]$$

$$= \frac{\pi}{2} \left(\frac{\pi}{6} + \frac{1}{4} \right) \text{ units}^3$$

Question 13

a i) $N = P + 100e^{kt}$
 $\frac{dN}{dt} = k(100e^{kt})$
 $= k(N - P)$

ii) $t=0$ $600 = P + 100e^{k \times 0}$
 $N=600$ $600 = P + 100$
 $P = 500$
 $\frac{dN}{dt} = 12$
 $\frac{dN}{dt} = k(100e^{kt})$
 $12 = 100k$
 $k = 0.12$

b)i) SHM $y = a - b \cos nt$
 a - centre of motion
 b - amplitude
 $\frac{2\pi}{n}$ - period

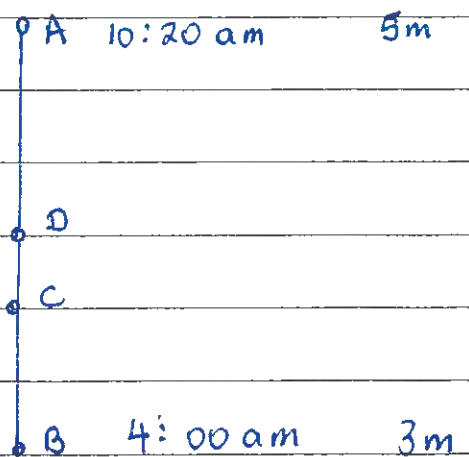
By inspection $a = 4$ $b = 1$

Time for one cycle is $2 \times 6\frac{1}{3}$ hours

$$\frac{2\pi}{n} = 12\frac{2}{3}$$

$$n = \frac{3\pi}{19}$$

$\therefore$ Water depth is given by $y = 4 - \cos \frac{3\pi t}{19}$



b ii)

$$y = 4.5$$
$$t = ?$$

$$4 - \cos \frac{3\pi t}{19} = 7.5$$

$$\cos \frac{3\pi t}{19} = -\frac{1}{2}$$

$$\frac{3\pi t}{19} = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$t = \frac{38}{9} \text{ hours (4h 13 min)}$$

$$t = \frac{76}{9} \text{ hours (8h 26 min)}$$

∴ Boat can be in the harbour between
8:13am and 12:26pm

c) i) $v = \frac{dx}{dt} = \frac{16-x^2}{x}$

$$\frac{dt}{dx} = \frac{x}{16-x^2}$$

$$\frac{dt}{dx} = -\frac{1}{2} \times \frac{-2x}{16-x^2}$$

$$t = -\frac{1}{2} \ln(16-x^2) + c$$

When $t=0$ $x=1$

$$0 = -\frac{1}{2} \ln 15 + c$$

$$c = \frac{1}{2} \ln 15$$

$$t = -\frac{1}{2} \ln \left(\frac{16-x^2}{15} \right)$$

$$-2t = \ln \left(\frac{16-x^2}{15} \right)$$

$$e^{-2t} = \frac{16 - x^2}{15}$$

$$15e^{-2t} = 16 - x^2$$

$$x^2 = 16 - 15e^{-2t}$$

$$x > 0 \text{ for } t = 0$$

$$x = \sqrt{16 - 15e^{-2t}}$$

cii)

As $t \rightarrow \infty$

$$e^{-2t} \rightarrow 0$$

$\therefore x$ approaches $\sqrt{16} = 4$

Particle moves right towards a limiting position 4m right of 0.

$$d) f(x) = \sin^{-1}(1-x) + \frac{\pi}{2}$$

$$\text{Domain } -1 \leq 1-x \leq 1$$

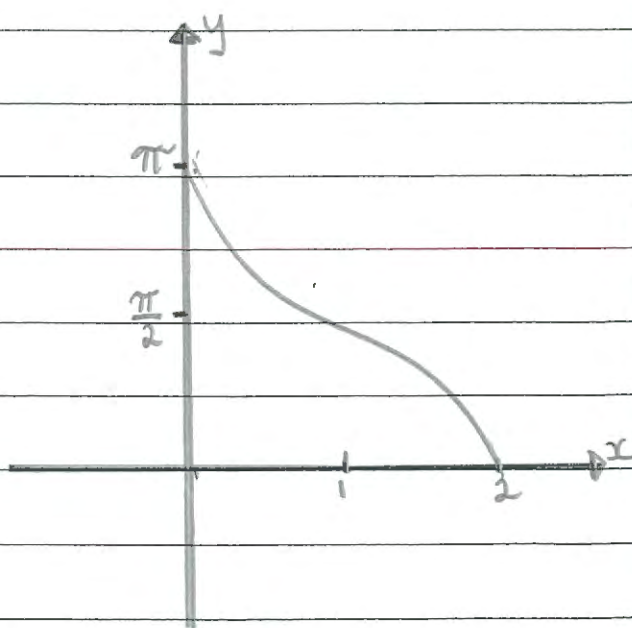
$$-1 \leq x-1 \leq 1$$

$$0 \leq x \leq 2$$

$$\text{Range } -\frac{\pi}{2} \leq \sin^{-1}(1-x) \leq \frac{\pi}{2}$$

$$0 \leq \sin^{-1}(1-x) + \frac{\pi}{2} \leq \pi$$

$$0 \leq f(x) \leq \pi$$



Question 4

a.i) The maximum height occurs when $y = 0$

$$y = 70 \sin \theta - 9.8t$$

$$0 = 70 \sin \theta - 9.8t$$

$$t = \frac{70 \sin \theta}{9.8}$$

$$\therefore y = \frac{70 \times 70 \sin^2 \theta}{9.8} - \frac{4.9 \times 70 \times 70 \times \sin^2 \theta}{9.8 \times 9.8}$$

$$= 500 \sin^2 \theta - 250 \sin^2 \theta$$

$$= 250 \sin^2 \theta$$

a.ii) $t = \frac{70 \sin \theta}{9.8}$

$$x = 70 \times \frac{70 \sin \theta}{9.8} \times \cos \theta$$

$$x = 500 \cos \theta \sin \theta$$

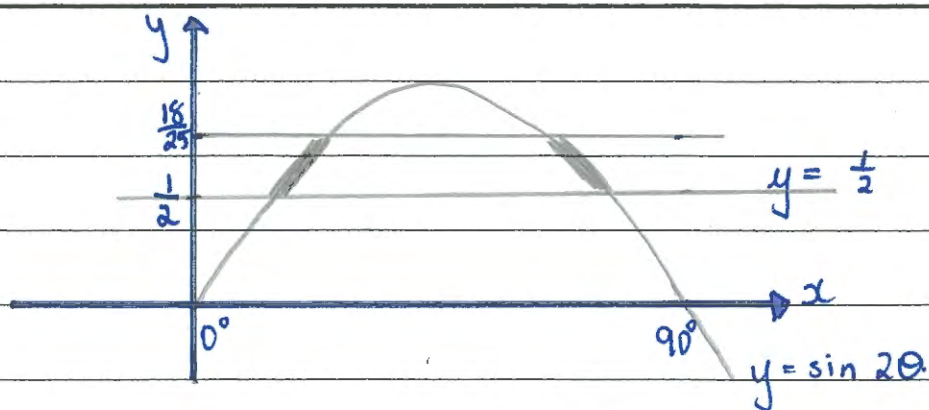
$$= 250 (2 \sin \theta \cos \theta)$$

$$= 250 \sin 2\theta$$

$\therefore$ the horizontal distance from O is $250 \sin 2\theta$ m

a.iii) We want $125 \leq 250 \sin 2\theta \leq 180$

$$\frac{1}{2} \leq \sin 2\theta \leq \frac{18}{25}$$



$$\frac{1}{2} \leq \sin 2\theta \leq \frac{18}{25}$$

$$30^\circ \leq 2\theta \leq 46^\circ$$

$$15^\circ \leq \theta \leq 23^\circ$$

$$134^\circ \leq 2\theta \leq 150^\circ$$

$$67^\circ \leq \theta \leq 75^\circ$$

for $y \geq 150^\circ$ and $0^\circ \leq \theta \leq 90^\circ$

$$250 \sin^2 \theta \geq 150^\circ$$

$$\sin^2 \theta \geq \frac{150}{250}$$

$$\sin^2 \theta \geq 0.6$$

$$\sin \theta \geq 0.774$$

$$\theta \geq 50.76$$

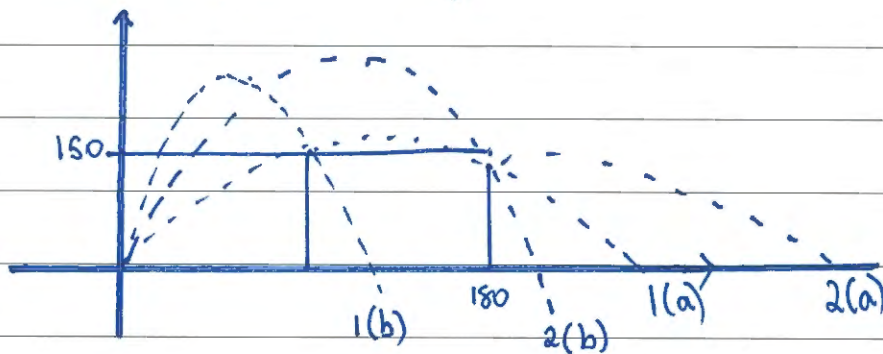
$$15^\circ \leq \theta \leq 23^\circ$$

$$67^\circ \leq \theta \leq 75^\circ$$

To satisfy both conditions

$$67^\circ \leq \theta \leq 75^\circ$$

Other method for 14 aiii)



$$y = x \tan \theta - \frac{x^2}{1000} - \frac{x^2}{1000} \tan^2 \theta$$

To pass through (125, 150)

$$150 = 125 \tan \theta - \frac{125^2}{1000} - \frac{125^2}{1000} \tan^2 \theta$$

$$5 \tan^2 \theta - 40 \tan \theta + 53 = 0$$

$$\tan \theta = 6.324 \text{ or } 1.676 \quad (\text{by quadratic function})$$

$$\theta = 81^\circ \text{ or } 59^\circ 11' \quad (\text{flight shown above})$$

answer 1(b)

answer 1(a)

To pass through (180, 150)

$$150 = 180 \tan \theta - \frac{180^2}{1000} - \frac{180^2}{1000} \tan^2 \theta$$

$$\therefore 162 \tan^2 \theta - 900 \tan \theta + 912 = 0$$

$$\therefore \tan \theta = 4.22 \text{ or } 1.33$$

$$\theta = 76^\circ 41' \text{ or } 53^\circ 4'$$

answer 2(b)

answer 2(a)

To denote in the required area, the rocket needs to be launched between 1(a) and 2(b) i.e. between $59^\circ 11'$ and $76^\circ 41'$

b i) $1 + x + x^2 + x^3 + \dots + x^n$

$r=1$ $S_n = \frac{1(x^{n+1} - 1)}{x - 1}$

$n+1$ terms

$$S_n = \frac{x^{n+1} - 1}{x - 1}$$

bii) Differentiating both sides

$$1 + x + x^2 + x^3 + \dots + x^n = \frac{x^{n+1} - 1}{x - 1}$$

$$1 + 2x + 3x^2 + \dots + nx^{n-1} = \frac{(x-1)(n+1)x^n - (x^{n+1} - 1) \cdot 1}{(x-1)^2}$$

Sub $x=2$ into both sides

$$1 + 2 \times 2 + 3 \times 2 + \dots + n \times 2^{n-1} = \frac{(2-1)(n+1)2^n - (2^{n+1} - 1) \cdot 1}{(2-1)^2}$$

$$= (n+1)2^n - 2^{n+1} + 1$$

$$= n2^n + 2^n - 2^{n+1} + 1$$

$$= n2^n + 2^n - 2 \cdot 2^n + 1$$

$$= 2^n(n+1-2) + 1$$

$$= 2^n(n-1) + 1$$

as required.

ci),

$$\sin \alpha = \frac{PG}{u}$$

$$\cos \alpha = \frac{BG}{u}$$

$$PG = u \sin \alpha$$

$$BG = u \cos \alpha$$

In $\triangle GBA$

$$AG^2 = BG^2 + 1^2 - 2 \times BG \cos 60^\circ$$

$$= BG^2 + 1^2 - 2 \times BG \times \frac{1}{2}$$

$$= BG^2 + 1 - BG$$

$$= (u \cos \alpha)^2 + 1 - (u \cos \alpha)$$

In $\triangle APG$

$$r^2 = PG^2 + AG^2$$

$$= (u \sin \alpha)^2 + (u^2 \cos^2 \alpha + 1 - u \cos \alpha)$$

$$= u^2 \sin^2 \alpha + u^2 \cos^2 \alpha + 1 - u \cos \alpha$$

$$= u^2 (\sin^2 \alpha + \cos^2 \alpha) + 1 - u \cos \alpha$$

$$r^2 = u^2 + 1 - u \cos \alpha$$

$$r = \sqrt{1 + u^2 - u \cos \alpha}$$

$$\text{c ii, } \frac{du}{dt} = 360 \quad t = 5 \text{ min or } \frac{1}{12} \text{ hour}$$

$$u = 360 \times \frac{1}{12} = 30 \text{ km}$$

$$r = \sqrt{1 + u^2 - 2u \cos \alpha}$$

$$= (1 + u^2 - 2u \cos \alpha)^{\frac{1}{2}}$$

$$\frac{dr}{du} = \frac{1}{2} (1 + u^2 - 2u \cos \alpha)^{-\frac{1}{2}} (2u - \cos \alpha)$$

$$= \frac{2u - \cos \alpha}{2\sqrt{1 + u^2 - 2u \cos \alpha}}$$

$$\text{As } u = 30$$

$$\frac{dr}{du} = \frac{2 \times 30 - \cos \alpha}{2\sqrt{1 + 30^2 - 30 \cos \alpha}}$$

$$= \frac{60 - \cos \alpha}{2\sqrt{901 - 30 \cos \alpha}}$$

$$\frac{dr}{dt} = \frac{dr}{du} \times \frac{du}{dt}$$

$$= \frac{60 - \cos \alpha}{2\sqrt{901 - 30 \cos \alpha}} \times 360$$

$$= 180 \left(\frac{60 - \cos \alpha}{\sqrt{901 - 30 \cos \alpha}} \right) \text{ km/h}$$

TRIAL HSC 2019 EXTENSION 1 – Markers comments

Question 11:

- a) $x \neq \frac{1}{4}$ needs
- b) Please don't divide by $\cos x$ as you lose half the answers and remember the correct quadrants in RADIANS.
- c) $\tan^{-1}\left(\frac{-3}{4}\right) = 4^{\text{th}}$ quadrant
- d) Learn the rule to differentiate inverse cosine
- e) Learn the correct reasons and state which triangle etc you are referring to. Most students had very poor setting out and reasoning.

Question 12:

- a) Please learn how to correctly set out mathematical induction. Use LHS = = RHS method working methodically down the page. Also, don't forget to add that k is a positive integer.
- b) Some students did not realise that you need to link the area of a circle to obtain $\frac{dA}{dr} = 2\pi r$ and substitute in the given information to find the value of r .
- c) (i) Show means do not skip steps. You must derive the gradient of the tangent at P and the gradient of the chord PQ first for one mark.
(ii) Poorly done question. You must always eliminate the parameter and show the locus in Cartesian form.
- d) Don't forget to change the bounds to be in terms of u . Silly mistakes when integrating $\frac{1}{u^2}$ lead to carry errors.
- e) Several students thought that $\frac{\pi}{12}$ is a larger fraction than $\frac{\pi}{4}$ and put the bounds in the wrong spot. Many stated that the integral of $\cos 2y$ is $-\frac{1}{2}\sin 2y$. You should know your basic integrals and if necessary, refer to the reference sheet to check and avoid silly mistakes with the signs.

Question 13:

- aii) about half of you didn't solve this correctly, "initially" means $t=0$ so no log involved.
- bi) small sketch or mention of amplitude needed to be made, not just assumption. n needed to be calculated with formula, centre of motion is 4. This with some explanation, achieved 2 marks.
- ci) one mark deducted if in final step no mention of initial condition.
- d) If not sure of shape of graph or where to start, make substitutions.